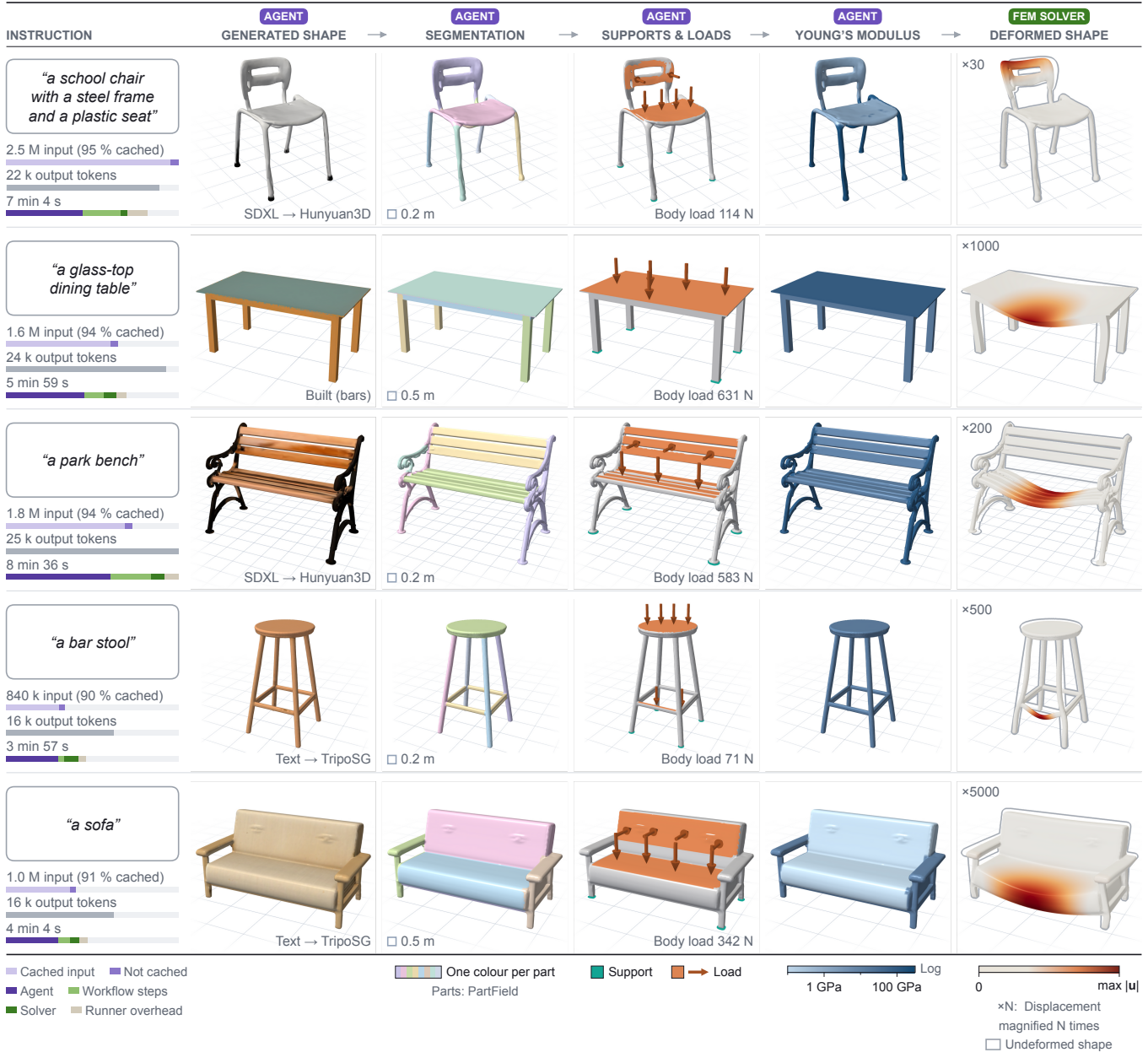

INSTRUCTION
AGENT
GENERATED SHAPE
AGENT
SEGMENTATION
AGENT
SUPPORTS & LOADS
AGENT
YOUNG'S MODULUS
FEM SOLVER
DEFORMED SHAPE
"a school chair with a steel frame and a plastic seat"
2.5 M input (95 % cached)
22 k output tokens
7 min 4 s
SDXL → Hunyuan3D
0.2 m
Body load 114 N
×30
"a glass-top dining table"
1.6 M input (94 % cached)
24 k output tokens
5 min 59 s
Built (bars)
0.5 m
Body load 631 N
×1000
"a park bench"
1.8 M input (94 % cached)
25 k output tokens
8 min 36 s
SDXL → Hunyuan3D
0.2 m
Body load 583 N
×200
"a bar stool"
840 k input (90 % cached)
16 k output tokens
3 min 57 s
Text → TripoSG
0.2 m
Body load 71 N
×500
"a sofa"
1.0 M input (91 % cached)
16 k output tokens
4 min 4 s
Text → TripoSG
0.5 m
Body load 342 N
×5000
Cached input
Not cached
Agent
Workflow steps
Solver
Runner overhead
One colour per part
Parts: PartField
Support
Load
Log
1 GPa
100 GPa
0
max |u|
×N: Displacement magnified N times
Undeformed shape